43. A statement followed by Conclusion-I and Conclusion-II is given below. You have to take the Statements to be true even if it seems to be at variance from the commonly known facts. Read all Conclusions and then decide which of the given Conclusion (s) logically follows/fallow from the Statement, disregarding the commonly known facts.

Statement : Some cats are almirahs. Some almirahs are chairs. All chairs are tables.

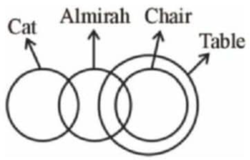
Conclusion-I : Certainly some almirahs are tables.

Conclusion-II : Some cats may not be chairs.

Which one the following is correct?

(a) Only Conclusion-I
(b) Only Conclusion-II
(c) Both Conclusion-I and Conclusion-II
(d) Neither Conclusion-I nor Conclusion-II

Ans. (c) :



Both Conclusion-I and Conclusion-II are correct.

44. The number 3798125P369 is divisible by 7. What is the value of the digit P?

(a) 1 (b) 6
(c) 7 (d) 9

Ans. (b) : 3798125P369

By using divisibility of 7

037 981 25P 369

$\Rightarrow (369 + 981) - (25\,P + 37)$

$\Rightarrow 1350 - 25\,P - 37$

$\Rightarrow 1313 - 25\,P$

by option, Put P = 6

$\Rightarrow 1313 - 256 \Rightarrow 1057$

$\because$ 1057, is completely divisible by 7.

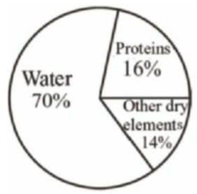
$\therefore \boxed{P = 6}$

BASIC NUMERICAL ABILITY

45. A pie diagram shows the percentage distribution of proteins, water and other dry elements in the human body. Given that proteins in the human body. Given that proteins correspond to 16% and water corresponds to 70%. If both proteins and the other dry elements correspond to p%, then what is the central angle of the sector representing p on the pie diagram?

(a) 54° (b) 96°
(c) 108° (d) 120°

Ans. (c) :



$\because$ Other dry elements %

$= [100 - (70 + 16)]\%$

$= (100 - 86)\%$

$= 14\%$

then, proteins and other dry elements $(P\%) = 16\% + 14\%$

$P\% = 30\%$

$\therefore$ Required Angle $= \frac{30}{100} \times 360^\circ$

$= 108^\circ$

46. Joseph visits the club on every 5th day, Harsh visits on every 24th day, while Sumit visits on every 9th day. If all three of them met at the club on a Sunday, then on which day will all three of them meet again?

(a) Monday (b) Wednesday
(c) Thursday (d) Sunday

Ans. (b) : $\because$ Joseph visits the club on every 5th day,
Harsh visits the club on every 24th day.
and Sumit visits the club on every 9th day.
then,
Required number of day meeting again
= the LCM of 5, 24 and 9.
$\because$ LCM = 360 Days

$\because$ Odd days = remainder of $\frac{360}{7}$

Odd days = 3
Hence, the day of again meeting = Sunday + 3
= Wednesday.

47. The difference between a 2-digit number and the number obtained by interchanging the positions of the digits is 54.

1. The sum of the two digits of the number can be determined only if the product of the two digits is known.
2. The difference between the two digits of the number can be determined.

Which of the above statements is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (b) : Let two digit number $= 10y + x$
The number obtained by interchanging the positions
$= 10x + y$
According to question
$(10y + x) - (10x + y) = 54$
$9y - 9x = 54$

$\boxed{y - x = 6}$

Considering the statements
From statement (1)
"The sum of the two digits of the number can be determined only if the product of the two digits is known."
$\therefore$ The statement is not correct due to "only if".
If both digits of the number is given, we conclude product of the two digits.
From statement (2)
"The difference between the two digits of the number can be determined."
Hence, Only statement (2) is correct.

48. X said to Y, "At the time of your birth I was twice as old as you are at present." If the present age of X is 42 years, then consider the following statements.

1. 8 years ago, the age of X was five times the age of Y.
2. After 14 years, the age of X would be two times the age of Y.

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b) :

Age/person	X	Y
The age, at the time of Birth of y	2a	0
Present age	3a	a

Let the present age of y = a
then the age of x at the time of birdh of y = 2a
and present age of x = 2a + a $\Rightarrow$ 3a
According to question
3a = 42
a = 14
Consider the statement (I)
$\because$ Present age of x = 42
and present age of y = 14
$(42-8) \gtreqless (14-8)\times 5$
$34 \gtreqless 6 \times 5$
$34 \gtreqless 30$
$\because$ $34 \neq 30$
Statement (1) is not correct
Considering the statement (2)
$(42+14) \gtreqless (14+14)\times 2$
$56 \gtreqless 56$
$\because$ 56 = 56
Statement (2) is correct
Hence, statement (2) is only correct.

49. If the price of an article is decreased by 20% and them the new price is increased by 25%, then what is the net change is the price?

(a) 0%
(b) 5% increase
(c) 5% decrease
(d) Cannot be determined due to insufficient data

Ans. (a) : Let the initial price = ₹ 100

After changing price $= 100 \times \frac{(100-20)}{100} \times \frac{(100+25)}{100}$

$= 100 \times \frac{80}{100} \times \frac{125}{100}$

= ₹ 100

Net change % $= \frac{(100-100)}{100} \times 100\%$

$= 0\%$

Price change % $= \left[-20 + 25 - \frac{20 \times 25}{100}\right]\%$

$= [+5 - 5]\% = 0\%$

50. When a certain numbers is multiplied by 7, the product entirely comprises ones only (1111....). What is the smallest such number?

(a) 15713 (b) 15723
(c) 15783 (d) 15873

Ans. (d) : By options:-
Option (a) $\Rightarrow$ 15713 × 7
= 109991
Option (b) $\Rightarrow$ 15723 × 7
= 110061
Option (c) $\Rightarrow$ 15783 × 7
= 110481
Option (d) $\Rightarrow$ 15873 × 7 = 111111
Hence, option d is correct for the given condition
$\therefore$ smallest number = 15873

51. A man completes 7/8 of a job in 21 days. How many more days will it take him to finish the job if quantum of work is further increased by 50%?

(a) 24 (b) 21
(c) 18 (d) 15

Ans. (d) : $\because$ Time taken to do $\frac{7}{8}$ of a work = 21 days.

$\therefore$ Time taken to complete the work $= 21 \times \frac{8}{7}$

= 24 days

and time taken to do 150% of a work

$= 24 \times 150\%$

$= 24 \times \frac{150}{100} = 36$ days

then, the number of more days = 36 – 21 $\Rightarrow$ 15 days

52. Using 2, 2, 3, 3, 3 as digits, how many distinct numbers greater than 30000 can be formed?

(a) 3 (b) 6
(c) 9 (d) 12